Unit 1, Lecture 1
Introduction to AI and Responsible AI

Today we will understand why AI systems need responsibility, fairness and human oversight before we trust their decisions.

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Why are we studying Responsible AI?

AI is no longer limited to laboratories; it is used in phones, banks, hospitals, classrooms and government systems.

When AI makes decisions, it can affect human opportunities, safety and dignity.

Therefore, our target is not only intelligent AI, but trustworthy AI. In this subject, we will learn how to build, evaluate and question AI systems responsibly.

Opening Question: Think Before Trusting AI

1. Prediction

If an AI predicts that a student may fail, should the teacher believe it blindly?

2. Decision

If an AI rejects a loan application, should the applicant know the reason?

3. Data

If an AI model is trained on biased data, can the output be fair?

4. Accountability

If an autonomous car makes a mistake, who is responsible?

AI Around Us: Students Already Use It

Search engines rank information for us.

YouTube, Instagram and Netflix recommend content.

Google Maps predicts routes and traffic.

Face unlock identifies users from images.

ChatGPT and Gemini generate text, code and explanations.

Banks use AI for fraud detection and credit scoring.

What is Artificial Intelligence?

Artificial Intelligence enables machines to perform tasks that normally require human intelligence.

These tasks include learning, reasoning, decision-making, perception, language understanding and prediction.

AI systems usually learn patterns from data and use those patterns to make decisions or generate outputs.

Important point: AI does not “understand” like humans; it processes data through mathematical models.

AI, ML, DL and Generative AI

Artificial Intelligence	Machine Learning	Deep Learning	Generative AI
Broad area: machines performing intelligent tasks →	Models learn patterns from data →	Neural networks learn complex representations →	Models create text, images, code, audio or video

Example flow: spam filter → image classifier → transformer model → ChatGPT-like assistant

How AI Learns: Data, Model and Prediction

Data: examples collected from the real world.

Model: mathematical function that learns patterns from data.

Training: process of adjusting the model using examples.

Prediction: model output for a new case.

Risk: if data is incomplete, biased or noisy, the model may learn wrong patterns.

Data		Model		Prediction
loan history exam marks images	→	learns pattern	→	approve/reject pass/fail class label

What is Responsible AI?

Responsible AI is the practice of designing, developing and deploying AI systems in a safe, fair, transparent and accountable manner.

It asks: Is the system useful? Is it fair? Is it explainable? Is it safe? Who is accountable?

Responsible AI combines technology, ethics, law, governance and human values.

The goal is to protect people while still using the power of AI.

What Can Go Wrong in AI Systems?

Bias: the model performs unfairly for certain groups.

Hallucination: the model gives confident but incorrect information.

Privacy leakage: sensitive data is exposed or misused.

Opacity: users cannot understand why a decision was made.

Automation bias: humans over-trust machine decisions.

Safety failure: the system acts incorrectly in high-risk situations.

Real-World AI Failures: Why We Need This Course

Chatbot hallucination

Incorrect but convincing answers can mislead users.

Hiring algorithm bias

Past hiring patterns may create unfair future screening.

Facial recognition error

Unequal accuracy can harm innocent people.

Deepfakes

Synthetic media can spread misinformation.

Core Principles of Responsible AI

Fairness	Avoid unjust discrimination.
Transparency	Make the system understandable.
Accountability	Assign responsibility for outcomes.
Privacy	Protect sensitive data.
Safety	Reduce harm and failure risks.

Ethics, Morality and Law: Basic Difference

Morality

Personal or social sense of right and wrong

Example: Should I submit AI-generated work without disclosure?

Ethics

Systematic principles for responsible conduct

Example: Should a model treat all user groups fairly?

Law

Rules enforced by institutions or government

Example: Data protection and privacy compliance

Responsible AI needs all three: human values, ethical reasoning and legal compliance.

Immediate and Long-Term Risks from AI

Immediate risks: misinformation, hallucination, privacy leakage, biased decisions, deepfakes and unsafe automation.

Long-term risks: loss of human control, concentration of power, large-scale job disruption and high-impact autonomous systems.

Responsible AI tries to identify risks early and reduce them before deployment.

Risk analysis is not only a technical issue; it is also a social responsibility.

How This Course Will Train You to Think

As a developer: build models carefully and test for fairness.

As a user: question AI outputs instead of accepting them blindly.

As a researcher: report limitations honestly and explain model behavior.

As a citizen: understand how AI affects society, privacy and public decision-making.

As an engineer: design AI systems that people can trust.